How Much Replication Is Enough? Balancing Novelty and Replication in a Truth-Driven Evolutionary Simulation

Kaitlyn Harper and Felix Schönbrodt

Department of Psychology, Ludwig-Maximilians-Universität München, Munich, Germany

Author Note

Kaitlyn Harper https://orcid.org/0000-0001-9597-872X

Felix Schönbrodt https://orcid.org/0000-0002-8282-3910

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Correspondence concerning this article should be addressed to Kaitlyn Harper, Email: research@kaitlynharper.com

Abstract

Scientific progress relies on a balance of original research, which furthers the boundaries of knowledge, and replication, which assures the quality of existing knowledge. In recent years, calls for increased replication have gained momentum, but replication requires resources that could otherwise support novel research. Here, we take a cultural-evolutionary approach to answer the question: what share of limited scientific resources should be allocated to original versus replication studies? We simulated a scientific ecosystem with a "utopian" selection criterion: whether research moves beliefs closer to the truth. Within this ecosystem, researchers differ in their tendency to pursue original or replication work, and more successful strategies (i.e., those which contribute most to truthful Bayesian information gain) are selected for over time. Under broadly realistic parameter assumptions, the model suggests an optimal replicator share of 5-32%, with 7-21% of published studies being replications, and replication success rates of 20-30% (all replications) and 72-88% (only published replications). Stronger publication bias, greater base null rates, and smaller original study sample sizes all increase the optimal replicator share. Our model may underestimate the optimal replicator share, because the model omits biases such as p-hacking that additional replication could help correct, or overestimate, because our model's simplistic choice of replication targets may be less efficient than more strategic ones. Ultimately, our simulations suggest that replication primarily evolves as a compensatory mechanism in response to distorted research environments, and that overall scientific progress is maximised when underlying issues such as publication bias are resolved, which would largely eliminate the need for replications.

Keywords: replication, metascience, cultural evolution, Bayesian inference, agent-based model

How Much Replication Is Enough? Balancing Novelty and Replication in a Truth-Driven Evolutionary Simulation

Scientific progress depends on a balance between original findings, which expand the boundaries of knowledge, and their replication[1], which ensures that these discoveries are reliable (Fisher, 1992; Popper, 2005). Over the past two decades, concerns about the replicability of scientific findings have intensified across many fields and large-scale replication efforts have revealed substantial rates of failure (Ioannidis, 2005; Nosek et al., 2022; Tyner et al., 2026). In response, the metascience community has increasingly called for greater investment in replication research to identify and correct errors in existing literature. However, critics have questioned whether calls for increased replications are practical, noting the absence of clear benchmarks for evaluating replication success rates (Rubin, 2024) as well as a tendency for broad metascientific reform proposals to be promoted without strong empirical grounding (Devezer et al., 2021). It is crucial to establish how much replication is actually necessary, because scientific resources - time, funding, and researcher effort - are finite. Allocating more resources to replication necessarily reduces the resources available for original studies, highlighting an inherent tradeoff between novelty and replication (Coles et al., 2018; Lewandowsky & Oberauer, 2020). These considerations point to a fundamental unanswered question: Given limited resources, what share of research effort should be allocated to replication?

A substantial body of work has addressed how to increase the number of replication studies (Koole & Lakens, 2012), how to prioritise which findings to replicate (Coles et al., 2018; Isager et al., 2025; Lewandowsky & Oberauer, 2020), how to design replication studies effectively (Hedges & Schauer, 2021; Pawel et al., 2023), and how to distribute replication efforts among researchers (Romero, 2020); yet there remains no empirical work on how resources should be distributed between original and replication research. As concerns about the efficiency and accountability of science intensify, how to best allocate scientific resources is becoming a core focus of discussion.

[1] With "replication" we refer to idealised direct/exact replications, where all aspects of the original study are kept constant except that the empirical data are from a new sample of the same population.

For instance, in a recent blog post, Dworkin (2025) proposed a system for assessing the value of replicating individual papers based on their age and citation counts. In this framework, the value of a replication lies in its potential to prevent further wasted research effort, weighed against the cost of conducting the replication itself. Using this approach, Dworkin estimated that 5–10% of existing NIH-funded papers would benefit from replication and subsequently 1–2% of NIH funding should be allocated to this purpose. However, this approach relies on strong assumptions - particularly regarding the base rate of true effects, which cannot be directly observed. The citation-based framework has also been critiqued for implying that papers with low expected future citation counts are not worth replicating. However, a classic finding may no longer attract many direct citations precisely because it has become part of the field's background knowledge, while many current theories, methods, or empirical claims may still depend on it being true (Buck & Dworkin, 2026). This approach would also overlook "sleeping beauty" papers, which receive little attention for years before suddenly becoming highly cited (Ke et al., 2015).

Modelling Replication in Academic Systems

The optimal balance between original and replication research is difficult to determine empirically, because the long-term effects of different allocation strategies cannot be directly observed in real scientific systems. Agent-based modelling can help by allowing us to simulate individual researchers and scientific knowledge over long timescales, capturing complex dynamics that are difficult to observe directly. A growing body of modelling work has investigated the role of replication within scientific communities. One line of work examines the epistemic role of replication itself. Population dynamics models show that replication can help distinguish true from false hypotheses, with its effectiveness depending on factors such as statistical power, base rates, and patterns of communication in the research community (McElreath & Smaldino, 2015). However, replication alone does not guarantee reliable knowledge accumulation: while replication can slow the spread of unreliable findings in systems that favour productivity over rigour, it cannot prevent it entirely (Smaldino & McElreath, 2016). Moreover, models suggest that high replicability is not necessarily aligned with truth, as systems optimised for replicable results do not always converge on correct conclusions (Devezer et al., 2019).

A second line of work focuses on the role of incentive structures in academic systems. In practice, scientific systems are not organised to optimise for truth, but are instead shaped by institutional incentives that reward novelty and productivity (Heesen, 2018). The "priority rule", for example, disproportionately rewards initial discoveries while offering little credit for subsequent verification (Merton, 1957; Romero, 2017). Publication bias, on the other hand, favours statistically significant results, leading to a "file drawer" of unpublished null findings (Rosenthal, 1979; Sterling, 1959). Agent-based models show that these incentives can systematically distort the scientific record by rewarding researchers who use poor methods and introducing a disproportionate amount of false positives into the literature (Higginson & Munafò, 2016; Smaldino & McElreath, 2016; Tiokhin et al., 2021). These incentives can also impact replication dynamics. Modelling by Ventura (2022) shows that reducing publication bias incentivises more researchers to engage in replication, and more replication can in turn incentivise more rigorous methods (as long as researchers are expecting their findings to be scrutinised).

Our Approach

Despite the insights outlined above, no existing model directly assesses the optimal share of research effort allocated to replication. We address this question using a model that manipulates the incentive structures of science and allows the optimal replicator share to emerge naturally. If current academic incentives can select for bad science, then aligning academic incentives with truthful scientific progress should, in principle, select for good science. Building on this idea, we modelled an evolutionary scientific ecosystem where selection pressures are aligned with epistemic goals. Rather than rewarding researchers for publication quantity, we modelled a scientific ecosystem with a "utopian" selection pressure, where researcher success is determined solely by the extent to which their published work contributes to truthful information gain (under a Bayesian framework of knowledge accumulation). Researchers conduct either original or replication studies, and strategies that most effectively move collective beliefs closer to the truth are more likely to persist and spread to new researchers over time. As there is a fixed number of researchers in the system, who can only research one effect at a time, time is the central resource constraint within the model. As a result, the composition of the scientific population will evolve toward an equilibrium that reflects the most epistemically efficient allocation of research effort (i.e., researcher time) among original and replication work. This approach allows us to directly address our central research question: Given limited resources, what share of research effort should be allocated to replication?

Methods

Overview

We investigated the optimal allocation of research effort between original research and replication using an agent-based model of cultural evolution in academia, implemented in R. The model allowed us to simulate how individual researcher behaviour, operating under selection pressure, shaped epistemic outcomes over time. Specifically, we wanted to know the replicator share that emerges under selection for truthful scientific progress. Within our model, selection is utopian: researchers' survival in academia relies on truth contribution, not the productivity- and novelty-based rewards that govern real academic careers. Researchers in our model neither p-hack nor fabricate data, thus observed effects depart from true values only through sampling variability - omitting other error sources that would likely raise the replicator shares we report (see Discussion). The academic system is otherwise intended to capture realistic aspects, including publication bias, so the replicator share that evolves can be read against plausible research conditions. We also varied three focal parameters across plausible ranges to assess their influence on the optimal allocation of research effort: the base null effect rate, the sample size of original studies, and the strength of publication bias. The model was preregistered using an ODD (Overview, Design concepts, and Details) model description (Grimm et al., 2006), available on Zenodo at https://zenodo.org/records/17465158; deviations from the preregistration are described in the Supplementary Material.

Figure 1 provides a visual overview of the model based on the ODD protocol (Szangolies et al., 2024). At each timestep, agents conduct either original or replication studies according to their research style. Studies take multiple timesteps to complete, contingent on sample size (i.e., larger sample sizes take longer to complete). When a study is completed, it is assigned a novelty contribution score, representing change in community beliefs (i.e., information gain), and a truth contribution score, representing change in community beliefs relative to the true effect. Both contribution scores follow a Bayesian belief updating framework and are explained further below. Publication bias then determines which studies enter the published literature, based on their statistical significance and novelty contribution. Only published studies update the actual community belief distribution for the relevant effect. At regular timestep intervals, researchers enter a career turnover phase, in which their retention or exit in the academic system depend on the cumulative truth contribution of their published studies during the preceding phase. Exited researchers are replaced with new researchers who inherit their research style from the remaining researchers, with a small chance of mutation. At the end of each timestep, the model records the current system state, including the distribution of research styles within the population and the overall scientific progress (i.e., the cumulative truth contribution of all studies). The following sections describe the model initialisation and the submodels depicted in Figure 1 in more detail.

Submodels

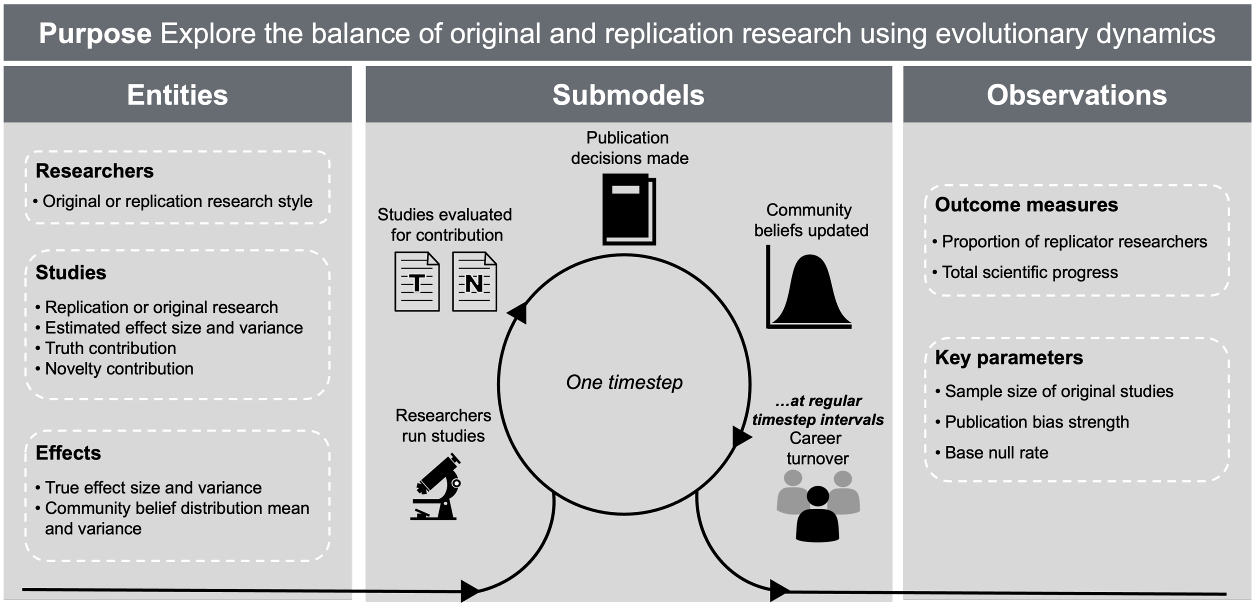
Initialisation

Each simulation begins with a population of 1,000 researchers, all with “original” research styles, and an effectively inexhaustible pool of effects. Each effect has a probability of being null set by the base null rate parameter. Non-null effects are drawn from a normal distribution with a mean of 0.3 and a standard deviation of 0.1. Each effect is also assigned a prior Bayesian community belief distribution, initialised as a normal distribution with a mean of 0 and a variance of 1. There are no existing studies at the beginning of a simulation.

Researchers run studies

Researcher time is the key limited resource in the model: only a fixed number of researchers are active at any time, each devotes their effort to a single study until it is finished, and there is no idle time between studies. Original researchers study effects that have not yet been

Figure 1
Model overview



Note. ODD = "T" and "N" indicate that studies were evaluated for their contribution to truth and novelty, respectively. Individual graphics were sourced from publicdomainvectors.org (CC0).

published. Replicators study published effects, prioritising those with the fewest published replications. Sample size per group for original studies is fixed within each simulation and varied across simulations (from 10 to 200 per group). Replication sample sizes are chosen dynamically to provide 80% power to detect the original reported effect when it was significant, or an effect of $d = 0.3$ when it was non-significant. Study duration increases by 0.1 timesteps for each observation per group, and original studies require one additional timestep for planning. All studies are simulated as independent-samples t-tests. When a study is conducted, its observed t statistic is drawn from a noncentral t distribution determined by the true effect size and the per-group sample size. Original studies use two-sided tests, whereas replications use one-sided tests in the direction of the original estimate.

Studies evaluated for contribution

Each completed study is evaluated according to the belief update it would produce if published. *Novelty contribution scores* capture how much the study would change community beliefs, regardless of whether the change is in the correct direction. *Truth contribution scores* capture how much the study would change community beliefs toward the true effect size (known to us, the modellers). Positive truth contributions mean the updated belief distribution places more weight on the truth than before the study, while negative values mean there is now less weight on the true value. Note that this framework also treats correctly learning that an effect is null as scientific progress. To be clear, this is a utopian measure that agent-based modelling allows us the opportunity to know and utilise, as in reality we have no access to the true effect size.

Let $p_{\text{prior}}(\theta)$ denote the existing community prior and $p_{\text{post}}(\theta)$ the posterior that would be obtained if the study were published. Studies are evaluated as such:

$$\text{Novelty contribution score} = D_{\text{KL}}\big(p_{\text{post}}(\theta) \,\|\, p_{\text{prior}}(\theta)\big)$$

$$\text{Truth contribution score} = \log p_{\text{post}}(\theta_{\text{true}}) - \log p_{\text{prior}}(\theta_{\text{true}})$$

Publication decisions made

Publication bias operates on two criteria: the novelty contribution of the study and the statistical significance of its finding (p-value). Publication bias follows two axioms: first, given the same level of novelty, significant results are more likely to be published than non-significant ones; and second, given the same status of significance, novel findings are more likely to be published than less novel ones. A study's probability of publication follows one of two logistic functions based on novelty and significance:

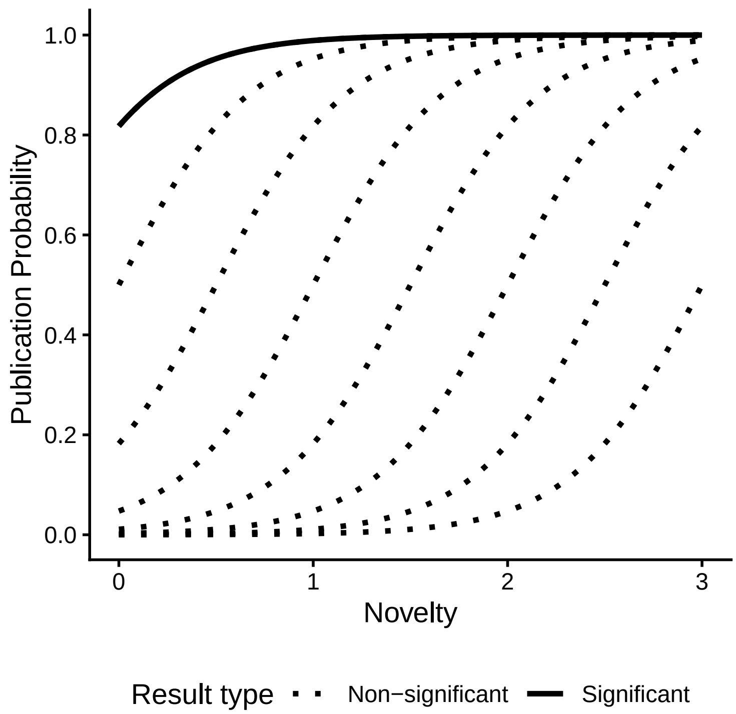
$$\Pr(\text{publish} \mid p < .05, N) = a_{\text{sig}} + \frac{1 - a_{\text{sig}}}{1 + \exp\left[-k_{\text{sig}}(N - m_{\text{sig}})\right]}$$

$$\Pr(\text{publish} \mid p \geq .05, N) = \frac{1}{1 + \exp\left[-k_{\text{nonsig}}(N - m_{\text{nonsig}})\right]}$$

where N is a study's novelty contribution, a_{sig} is the lower asymptote for significant results, m terms are logistic midpoints, and k terms are logistic steepness parameters.

Publication bias is operationalised through the m_{nonsig} term: higher values of publication bias mean a lower baseline probability that a non-significant study is published, and a higher novelty threshold for such a study to be published despite its non-significance (visualised in Figure 2). Note that the publication bias function applies both to original and replication studies in the focal simulations.

Figure 2
Publication bias functions



Note. Multiple logistic curves for non-significant findings are presented to visualise the range of publication bias conditions, with the m_{nonsig} term ranging from -0.5 to 3. At -0.5, non-significant and significant findings have the same probability of publication for a given level of novelty, so the two curves overlap and only the significant curve is visible. At 3, the probability of publishing a non-significant finding is essentially zero unless novelty is extremely high.

Community beliefs updated

Published studies update the community belief distribution for the relevant effect using normal-normal Bayesian updating, where y represents the study estimate and its uncertainty. The posterior then becomes the prior for the next published study of that effect.

$$p_{\text{post}}(\theta \mid y) = p(y \mid \theta)\, p_{\text{prior}}(\theta)$$

Career turnover

Career turnovers occur every 35 timesteps, allowing researchers to complete multiple studies under most sample-size conditions, while preserving the substantial time cost of very large studies. Researchers are ranked by the summed truth contribution of the studies they published during that period. The lowest-ranked half of agents exit academia, while the remaining agents are promoted, meaning they are allowed to continue in the academic system. Vacant positions are then filled by new researchers, who inherit their research styles from existing values in the active academic population, with a small chance of mutation (2% probability of switching from one style to the other) to capture imperfect transmission, individual variability, and innovation. The active population therefore remains fixed at 1,000 while the replicator share evolves over time.

Simulation Design and Outcomes

We summarise two main simulation end-state outcomes: total scientific progress and the replicator share. Total scientific progress measures how much the scientific community has learned by the end of the simulation, defined as the summed log-density gain at the true effect values from the uninformed prior to the final posterior. That is, it measures how much more strongly the scientific community comes to believe in the true effect values compared with where it started. The replicator share is the percentage of active researchers with a replication research style during the final 50 timesteps of the simulation. Because researchers work continuously on one study at a time, this is also the share of researcher time - that is, of research effort - allocated to replication. Thus, we can use the replicator share as a direct measure of how limited scientific resources are allocated between original and replication research.

First, we estimated the stochastic variability of the model by running one fixed parameter condition with 1,000 different random seeds, providing a benchmark for the amount of variation expected in outcomes from simulation dynamics alone. Then, the focal parameter sweep characterises the model's behaviour across three focal parameters: the base null rate, publication bias strength, and original study sample size. We used Latin hypercube sampling to select 10,000 combinations of base null rate (ranging 0 to 1), publication bias strength (ranging -0.5 to 3), and original study sample size (ranging 10 to 200). This approach spreads combinations evenly across the full parameter space, allowing us to examine persistent patterns rather than varying one parameter at a time. We examine the marginal effects and interactions between the focal parameters and highlight conditions with realistic parameter values.

To assess the stability of our results, we compared the resulting patterns across smaller independent batches of parameter combinations. We also conducted additional parameter sweeps in which certain non-focal parameters were changed to represent plausible alternative scenarios (e.g., what if true effects were larger). These additional results, as well as preregistration deviations, publication bias calibration, and extended methods with all default parameter values are reported in the Supplementary Material.

Results

Stochastic Variability

Before interpreting the main results, we first estimated the degree of stochastic variability produced by the simulation under fixed parameter values. To do this, we set default parameters and reran the model 1,000 times with different random seeds. This provides a baseline sense of how much variation in the outcomes should be expected from stochastic simulation dynamics alone, rather than from changes in model parameters. Across these repeated runs, the mean replicator share was 8.4%, with a standard deviation of 2.3% and a standard error of 0.07%. As a practical interpretive guideline for the single simulation run conducted at each parameter condition, we treat differences within approximately two standard deviations (5%) as too small to interpret confidently as meaningful model differences. This threshold is not intended as a formal hypothesis test, but as a rule of thumb for avoiding overinterpretation of small differences between parameter conditions.

Focal Parameter Sweep: Marginal Effects

Figure 3 shows the marginal effects of each focal parameter, averaged across the other two. Total scientific progress decreases as the base null rate increases, because more truly null effects create more opportunities for false positives to enter the literature. Publication bias has a much stronger negative effect: as it increases, scientific progress declines sharply, because true null results are less likely to be published and the literature becomes increasingly dominated by false positives. Original study sample size has a non-linear effect. Larger original studies initially improve scientific progress, but these gains level off (in our model, this occurs at ~50 observations per group for independent *t*-tests); beyond this point, additional observations are better spent studying new effects rather than increasing certainty about already-studied effects.

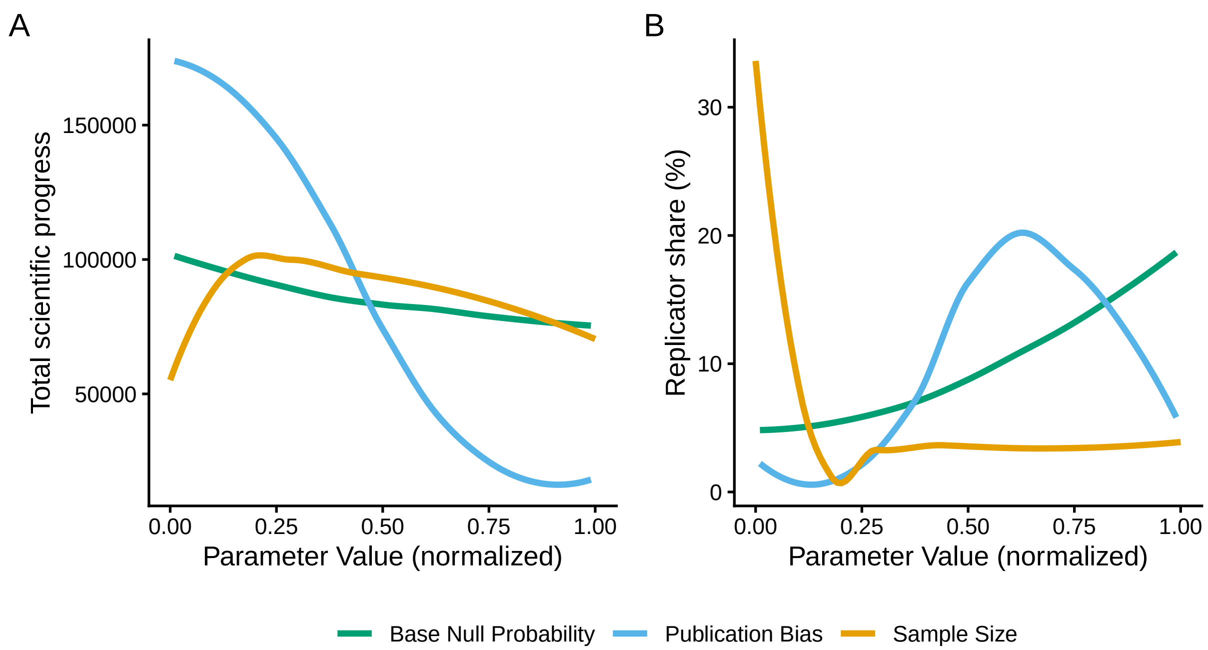
The evolved replicator share largely mirrors these patterns, such that more replication evolves in response to environments where less scientific progress is being made. More replication evolves when the base null rate is higher, consistent with replication becoming more valuable when the literature is more vulnerable to false positives. As original study sample size increases, replication initially declines because original studies become more informative and there is less need for replication. Greater publication bias also initially comes with more evolution of replication, but only up to a point. When publication bias is very strong - remember that it applies both to original and replication studies - even replication efforts with a high truth contribution are unlikely to enter the literature, and consequently fail to improve researchers' fitness and are no longer favoured.

Focal Parameter Sweep: Interaction Effects and Realistic Conditions

To complement the reported marginal patterns, we also show a more detailed breakdown of the focal sweep results in Figure 4, highlighting the results space corresponding to a realistic range of parameter values. Specifically, we treat the realistic range as a medium (0.33–0.66) to high (0.66–1.00) base null rate (Ioannidis, 2005), original study sample sizes of approximately 20–40 participants per group, and publication bias values between 2 and 3, based on the calibration analysis reported in the Supplementary Material. We chose the sample size range to be

Figure 3

Marginal effects of publication bias strength, base null rate, and original study sample size on (A) the total scientific progress at the end of the simulation and (B) the replicator share in the final population



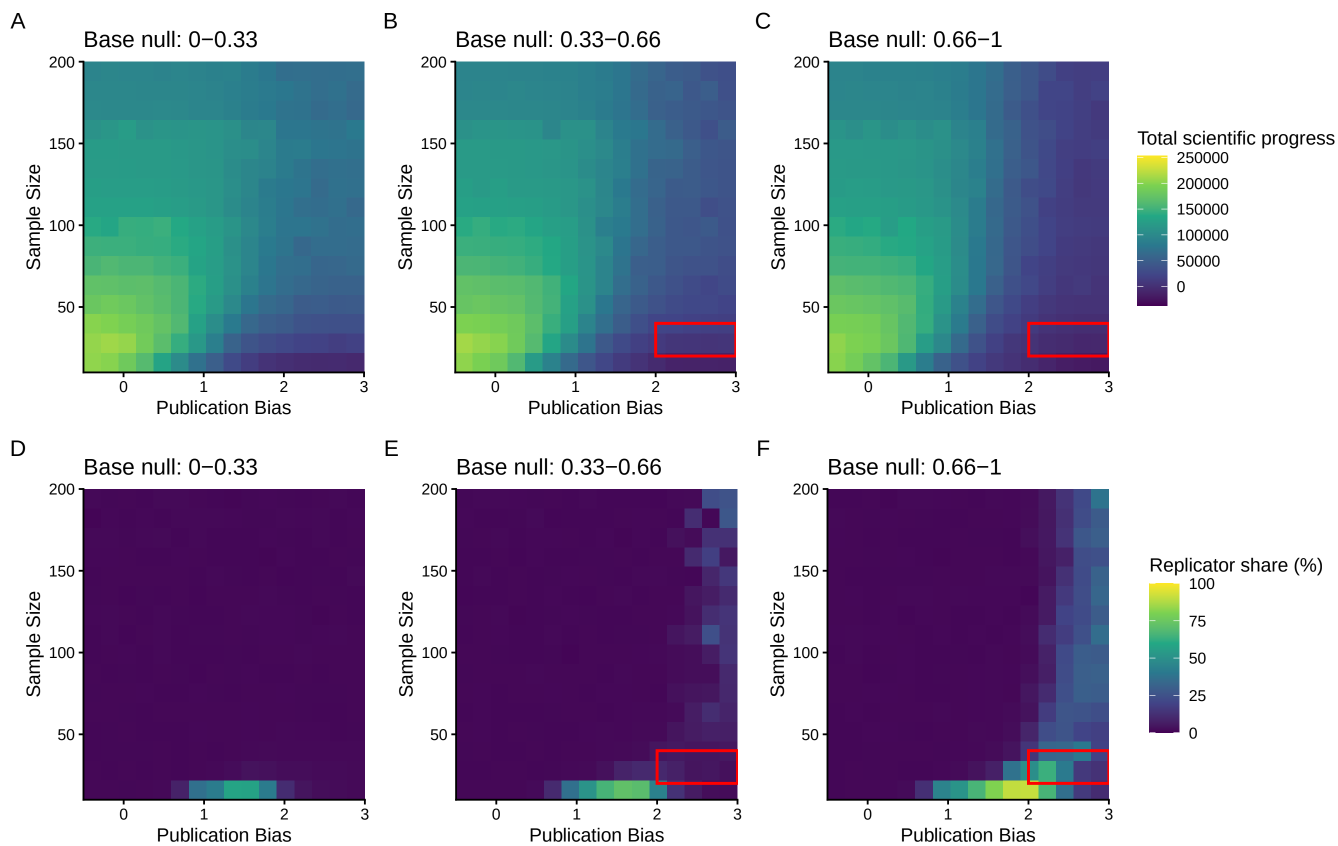
Note. The x-axis represents the full possible range of each parameter as follows. Sample size: 10 to 200, base null rate: 0 to 1, publication bias: -0.5 to 3.

broadly representative of historical sample sizes in psychology (Bakker et al., 2025; Fraley & Vazire, 2014; Marszalek et al., 2011), which feels relevant given the strong metascientific focus in this field. Although sample sizes have increased in more recent work (Bakker et al., 2025; Fraley et al., 2022), we emphasise the smaller historical values because they more closely reflect the existing literature base that replication would apply to.

In this realistic parameter region, the mean evolved replicator share ranged from 5% in the mid base null rate conditions to 32% in the high base null rate conditions. Thus, when null effects are common, original studies use relatively small sample sizes, and publication bias is fairly strong, the system needs a modest to substantial amount of replication to correct misleading findings and keep making truthful scientific progress. Additionally, 7% of published studies were replications in the mid base null rate conditions, compared with 21% in the high conditions. When all replication attempts were included (i.e., published and unpublished), mean replication success rates were 30% in the mid base null rate conditions and 20% in the high conditions; when only published replications were considered, they were 88% and 72%, respectively. We considered a replication to be successful using the simple significance criteria: that the significance and direction of the effect were consistent (Muradchanian et al., 2021).

Figure 4

Heatmaps of publication bias strength and original study sample size on (A) total scientific progress at the end of the simulation and (B) the replicator share in the final population, stratified by three bins of base null rate



Note. Each heatmap shows the joint effect of publication bias strength (horizontal axis) and original study sample size (vertical axis) within one base null rate bin; columns correspond to three equal-width bins of base null rate (0–0.33, 0.33–0.66, and 0.66–1). Red boxes highlight realistic parameter conditions.

Discussion

The replication crisis has prompted widespread calls for more replication research, but it has also raised a more difficult question: how much of science's limited resources should be allocated to replication rather than original research? Across our simulations, the share of research effort allocated to replication evolved in response to poorer epistemic conditions. When original findings are less reliable, replication becomes more valuable: higher base null rates, smaller original study sample sizes, and stronger publication bias all tended to increase the evolved replicator share. However, this adaptive response has limits. When publication bias becomes sufficiently strong, even informative replications may fail to enter the published record, weakening the corrective role that replication is meant to play. We also observed diminishing returns to larger sample sizes in original studies: beyond a certain point, concentrating more observations into each original study can reduce overall scientific progress, because those same resources may be more valuable when spread across a larger number of effects.

How much replication is enough?

Within the region of the parameter space which we take to reflect more realistic levels of publication bias, sample size, and base null rate, the evolved replicator share was 5%–32%. This should not be interpreted as the percentage of published articles that should be replications. Rather, in the context of the model, it refers to the share of research effort, specifically researcher time, allocated to replication that was optimal for truthful information gain. Under these conditions, replications accounted for 7%–21% of published studies. This published replication rate should not itself be treated as an optimal target: researcher style (original or replicator) is selected on directly for its contribution to truthful information gain, whereas the proportion of published studies that are replications emerges from the dynamics between researcher behaviour and the scientific environment. The same published replication rate could therefore correspond to very different levels of scientific progress under different conditions.

We should be cautious about treating any single value from our model as a prescription, since several modelling assumptions could shift the optimal allocation of scientific resources.

Most importantly, sampling error is the only source of study error in the model: agents do not p-hack, introduce researcher bias, or fabricate data. In reality, a non-zero level of these additional distortions is present, which would likely increase the amount of replication needed as a corrective mechanism, assuming that replications are conducted with more rigorous methods. The impact of such error sources is difficult to estimate, but simulations suggest that questionable research practices can substantially affect false positive rates (Stefan & Schönbrodt, 2023). At the same time, our model selects which studies to replicate relatively simply, whereas real-world replication efforts can be strategically directed toward influential, surprising, or methodologically important findings, and designed to maximise their evidential value (Coles et al., 2018; Hedges & Schauer, 2021; Isager et al., 2025; Lewandowsky & Oberauer, 2020; Pawel et al., 2023). More targeted and better-designed replications could therefore make each replication attempt more informative, potentially reducing the total amount needed. For this reason, we emphasise the broader pattern of results over the optimal allocation of resources estimated under our specific parameter settings.

Our model may also speak to debates about what replication success rate science should aim for (Rubin, n.d.). It may seem obvious that higher replication success is better: in an ideal scientific system where original studies and replications both accurately capture true effects, we would expect close correspondence between them. However, Devezer et al. (2019) demonstrate with modelling that this does not mean that a high replication success rate is, by itself, evidence of a healthy scientific system. A 100% replication success rate may emerge under perfect science, but it could also emerge for less desirable reasons, such as when the same biases affect both original studies and replications. Our model illustrates this point. In the realistic parameter conditions, replication success was much higher among published replications (72%–88%) than among all replications (20%–30%), demonstrating how publication bias can obscure the true success rate of replication. At the same time, the low success rate among all replications in these conditions should not be interpreted as a desirable target, even though it emerges in a model where replication evolves adaptively to improve scientific progress. Rather, it reflects the fact that replications are correctly identifying problems in a distorted published literature. For the real-world literature that already exists, the more immediate concern should be whether replication success rates accurately reflect its reliability, since we cannot retroactively improve this research. Going forward, replication success can also be improved by strengthening the quality of original studies themselves. The goal, then, should not be to maximise replication success rates directly per se, but to promote conditions under which high replication success would genuinely indicate that original studies and replications are both accurately estimating effects.

Publication Bias

Our results demonstrate the well-established finding that publication bias can strongly distort the scientific record. This aligns with previous modelling showing that selectively publishing positive results can allow false claims to become "canonised" as scientific facts (Nissen et al., 2016), and contrasts with accounts suggesting that publication bias may be less damaging than often assumed (Rubin, 2025). In our model, publication bias was not a minor imperfection in the system, but one of the main forces shaping whether scientific communities moved toward or even away from the truth. In this sense, replication may help treat the symptoms of a biased literature, but reducing publication bias would address the disease itself.

However, determining that publication bias is harmful does not by itself establish the best way forward. One obvious solution would be to eliminate publication bias entirely, but beyond the practical difficulty of changing publication systems, a "publish everything" approach may raise its own challenges. Nelson et al. (2012) argue that some work may belong in the "file drawer" and that publishing all papers can create a "cluttered office" problem: even if more information is technically available, it may become harder for researchers to identify the work that matters among many low-quality or uninformative papers. Zollman (2009) simulated journal behaviour and found that the optimal publication strategy involved an initial quality screening followed by a degree of randomness in what is published. De Winter and Happee (2013) argued that selective publication could sometimes improve meta-analytic estimation over a publish everything approach at a fixed number of published studies. However, this conclusion depends heavily on what is counted as the cost of science. Van Assen et al. (2014) showed that when unpublished studies are counted as expended scientific resources rather than treated as if they never existed, publishing everything is more efficient, especially when the true effect is null.

Some modelling assumptions limit our ability to contribute to this debate. We do not model low-quality studies, theoretically uninteresting research questions, or limits on researchers' ability to find and integrate evidence. Instead, all studies are assumed to be methodologically sound, and every published finding is automatically incorporated into the community's beliefs. In this way the community beliefs in our model are more like an idealised, continuously updated meta-analysis. McDiarmid et al. (2021) found that psychologists do revise their beliefs about effect sizes after learning the results of replication studies, although less than Bayesian modelling suggested they should. This supports our assumptions about how published replication evidence shapes beliefs, but also emphasises that evidence being available is not the same as evidence being fully integrated. Another assumption within our truthful information gain framework is that null results are not clutter: they are useful evidence that helps the community learn when an effect is small or absent. Under such assumptions, we find that a "publish everything" approach contributes the most to scientific progress.

The implication is not that every study should receive equal attention, but that results should not be hidden simply because they are non-significant. The challenge then is to make all results available while developing systems of filtering and synthesis that prioritise quality, relevance, and cumulative value rather than statistical significance. Proposals for making findings available regardless of outcome include Registered Reports (Scheel et al., 2021), journals devoted to non-significant or null results, and preprint or micropublication formats that make otherwise file-drawered work findable. For replications specifically, proposals include dedicated replication journals (Reed et al., 2025) and a "Pottery Barn rule" for science - "you break it, you buy it" - under which journals take responsibility for publishing credible replications of findings they originally published (Srivastava, 2012).

Assumptions and Limitations

Scientific models do not need to reproduce an academic system in full detail to be useful. Their value lies in making assumptions explicit and reducing complex systems to core mechanisms, allowing us to explore dynamics and consequences that are difficult to reason through verbally or observe in real life. The simplifications are not incidental, but central to the purpose of the model. In the present study, we make several such assumptions to isolate the relationship between replication, publication bias, and scientific progress. Beyond the assumptions already discussed in relation to our findings, the model also uses a highly simplified representation of effects: effects do not build on one another, are not embedded within broader theoretical structures, and are not linked by conceptual similarity. Researchers therefore do not choose hypotheses on the basis of theory, prior findings, or related lines of evidence, and replications are direct rather than conceptual. These simplifications allow us to focus cleanly on our research question regarding how to distribute resources among original and replication studies, but they also mean that the model should be understood as a baseline account rather than a complete representation of scientific practice.

Further assumptions come from our use of Bayesian updating to quantify scientific progress. This is a useful feature of the model because it provides an established framework for representing how evidence changes beliefs, however, it is not assumption-free: it implicitly determines how uncertainty reduction is epistemically valued. One such assumption concerns null effects. In our model, becoming more certain that an effect is null contributes to scientific progress in the same way as becoming more certain about a non-null effect. In reality, however, learning about null effects may not be highly valued, and may depend on the aims of a particular field of research. A second, related assumption concerns the balance between exploration and exploitation: whether finite scientific resources are better used to learn a little about many effects or a great deal about fewer effects. Our model does not explicitly specify a preferred balance between breadth and depth. Instead, this tradeoff is handled implicitly by the Bayesian scientific progress measure, which rewards both learning about previously uncertain effects and reducing uncertainty around effects that have already been studied. Consequently, the same overall amount of scientific progress can arise from quite different distributions of knowledge, and the particular way progress is measured may influence which allocation of research effort is favoured at equilibrium.

Our sample size results provide a concrete example of this tradeoff. The apparent advantage of modest original study sample sizes reflects a balance between making individual studies sufficiently informative to produce meaningful belief updating and preserving enough resources to investigate a broader range of effects. This balance is not chosen explicitly but instead emerges from the Bayesian updating framework. Related work shows that optimal resource allocation can depend strongly on the criterion used to evaluate scientific success. For example, under incentives for significance rather than truth, several smaller studies may outperform a single larger study (Bakker et al., 2012). We intend to use our model to further investigate the broader problem of how scientific systems balance exploration and exploitation: when to spread effort across many uncertain possibilities, when to concentrate effort on fewer effects, and how different measures of progress implicitly balance that tradeoff.

Conclusion

Across conditions, our model suggests that replication is most useful as a crutch in response to poor epistemic conditions. It can help when the published literature is distorted, but that help is limited when publication bias also suppresses replications themselves. In an ideal system where publication bias and related distortions were removed, replication would be less necessary, and in some model conditions not necessary at all. Our model highlights that the value of replication is not fixed, but depends on the extent to which the literature is distorted and whether scientific incentives reward work that improves truthful information gain. This points beyond the question of how much to invest in replication, toward the harder question of how institutions, journals, and funders might better recognise epistemically valuable work.

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